Data Dictionary — Image Classification (CNN) Dataset

Recommended Dataset: Fashion-MNIST

Fashion-MNIST is a drop-in replacement for the original MNIST digit dataset, with 70,000 grayscale images of clothing items in 10 classes. It is available directly in Keras/TensorFlow.

Dataset Overview

Attribute	Description
Source	`tf.keras.datasets.fashion_mnist.load_data()` or [Fashion-MNIST on GitHub](https://github.com/zalandoresearch/fashion-mnist)
Image shape	(28, 28) — height 28 pixels, width 28 pixels
Channels	1 (grayscale)
Pixel range	0–255 (integer); normalize to 0–1 (float) for training
Training samples	60,000
Test samples	10,000
Number of classes	10

Class Labels (Fashion-MNIST)

Label (int)	Class name
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat
5	Sandal
6	Shirt
7	Sneaker
8	Bag
9	Ankle boot

Preprocessing Notes

- Normalization: Divide pixel values by 255.0 so inputs are in [0, 1].
- Shape for Keras: Use shape `(28, 28, 1)` if the model expects a channel dimension, or `(28, 28)` and let Keras expand.
- Optional: Resize to a larger size (e.g., 32×32) if you want to experiment with more parameters; document any resizing.
- Data augmentation: Apply only to training data (e.g., rotation, width/height shift, horizontal flip). Do not augment test/validation data.

Alternative Datasets

- CIFAR-10: 32×32 RGB images, 10 classes (airplane, car, bird, etc.). Load via `tf.keras.datasets.cifar10.load_data()`.

- Cats vs Dogs: Binary classification; available on Kaggle or TensorFlow Datasets. Requires download and folder structure (train/cats, train/dogs).

Use the data dictionary for the dataset you actually use and note any extra columns or folder structure in your report.